

walking by a restaurant and looking through the glasses to check whether the menu is to his liking or not, so that he may decide whether to go in or not.

Meta 2 launched by Meta Vision in 2014, enables you to create holographic apps, tools and experiences. According to a tester, "Objects in the field-of-view behave like things in the real world." Priced at US\$ 949, the experience is nothing short of a fantasy. It allows fantasy objects to be lifted and placed in different locations just like moving furniture around the house. You can also experience walking around a globe and seeing it from different angles. Though there seem to be some issues with gesture recognition, this could very well lead the way to some truly interesting experiences in the future.

Laster SeeThru, however, takes a different approach. The headset does not have a camera to avoid issues regarding invasion of privacy. It works on its own series of location and GPS to get things done. The device is targeted at adventure sports and helps in navigation. Available for US\$ 399, the device connects with your mobile device to allow contacts access from phone, localisation and navigation.

And then we have the rumours. Known for their quality of service and beauty of design, Apple is also expected to enter the VR market. As per a report by Bloomberg, they have gone on to discuss the project with their hardware suppliers and have ordered near-eye displays for internal testing.

Rumours also indicate that an Apple AR headset would connect to an iPhone. These go on to suggest that secret teams are working on iGlasses, which should hit the market sometime by 2018. If the

quality of the current products from Apple is any consideration, the AR or VR device is expected to be a premium product.

However, one thing that sticks out is that, with VR or AR, need for displaying HD content is on the rise. So how are graphic processors keeping up?

What makes virtuality tick

Current headsets provide an updated view of the world every 11.1 milliseconds. Their graphic card



Meta 2 from Meta Vision

effectively generates a frame rate of about 90fps. This calls for improving the frame rate and keeping up with the high frame rate requirements. If the next frame is not ready in time, the perspective is adjusted based on the trajectory to cover up for the lag. Asynchronous Space Warp is a frame-rate smoothing technique that almost halves the CPU/GPU time required to produce nearly the same output from the same content. It enables users to run Oculus Rift headsets on lower specification hardware. Asynchronous Space Warp Vive is quite helpful with low-end GPUs that are incapable of generating frames at the required speed.

The currently top-of-the-line GPUs include Strix GTX 10 series processors and RX 480. Strix GTX based on Nvidia's Maxwell archi-

ture has undergone quite a few improvements. GTX 1060, 1070 and 1080 are improvements over their earlier 9 series processors with improved VR performance. While the 9 series delivers decent experiences overall, the 10 series depends on time-warping for higher graphic demanding games. RX 480, based on the third-generation Graphics Core Next architecture from AMD, is another improvement over the earlier 390 series processors.

There has also been reduction in the requirements for VR. Last year saw the processing requirements of Rift reduce from Nvidia GTX 970 or AMD 290, Intel i5-4590 and 8GB RAM to i3-6100 CPU with GTX 960 GPU or AMD FX-4350 with Radeon RX 470. This has, however, been in large due to Asynchronous Space Warp.

Looking purely at graphics processing, AMD Radeon RX480 remains the most cost-effective solution for VR experiences. It manages to generate about 62fps, whereas the alternative Nvidia's

GTX 1080 generates about 59fps. Looks like a long way to go for processors to hit the ideal 90fps mark!

The oxymoron goes on

Rumors or not, VR is a major focus area for manufacturers going ahead. Its popularity is evident from an experience I had last week. While travelling, I heard a knock on my car window. Thinking it would be another hawk selling stuff I did not need, I brushed him off. When the vehicle moved, the rear-view mirror showed a person holding three to four VR headset boxes. Those might have been knockoffs, but the idea that VR has reached the streets is heartening. **EFY**